

Download SQL Server Data Tools (SSDT) for Visual Studio

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APPLIES TO:  SQL Server  Azure SQL Database  Azure Synapse Analytics (SQL Data Warehouse)  Parallel Data Warehouse

SQL Server Data Tools (SSDT) is a modern development tool for building SQL Server relational databases, databases in Azure SQL, Analysis Services (AS) data models, Integration Services (IS) packages, and Reporting Services (RS) reports. With SSDT, you can design and deploy any SQL Server content type with the same ease as you would develop an application in Visual Studio.

SSDT for Visual Studio 2019

Changes in SSDT for Visual Studio 2019

The core SSDT functionality to create database projects has remained integral to Visual Studio.

With Visual Studio 2019, the required functionality to enable Analysis Services, Integration Services, and Reporting Services projects has moved into the respective Visual Studio (VSIX) extensions only.

Note

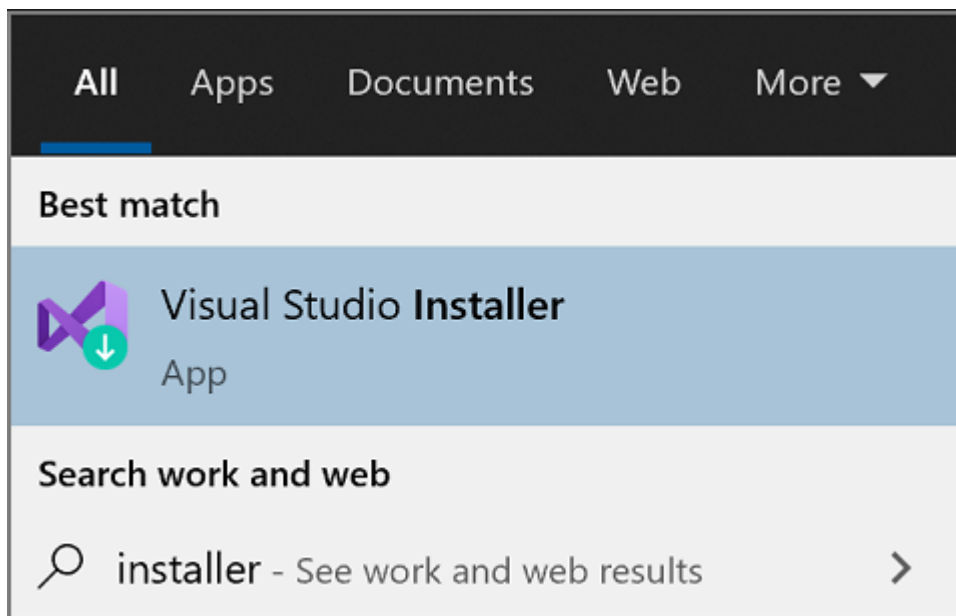
There's no SSDT standalone installer for Visual Studio 2019.

Install SSDT with Visual Studio 2019

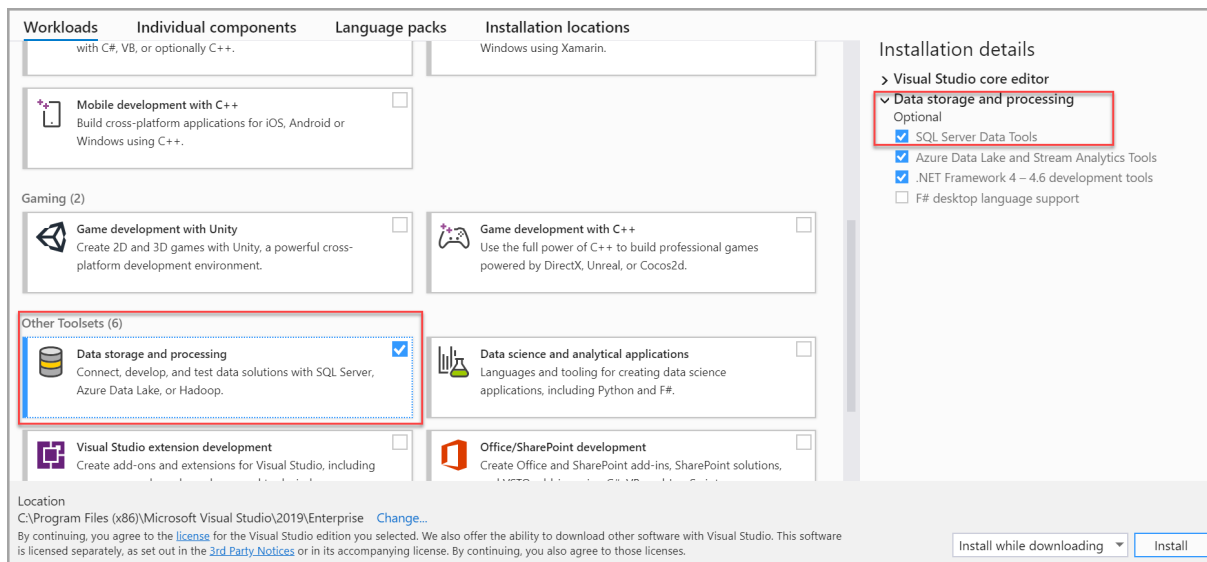
If [Visual Studio 2019](#) is already installed, you can edit the list of workloads to include SSDT. If you don't have Visual Studio 2019 installed, then you can download and install [Visual Studio 2019 Community](#).

To modify the installed Visual Studio workloads to include SSDT, use the Visual Studio Installer.

1. Launch the Visual Studio Installer. In the Windows Start menu, you can search for "installer".



2. In the installer, select for the edition of Visual Studio that you want to add SSDT to, and then choose **Modify**.
3. Select **SQL Server Data Tools** under **Data storage and processing** in the list of workloads.



For Analysis Services, Integration Services, or Reporting Services projects, you can install the appropriate [extensions](#) from within Visual Studio with **Extensions > Manage Extensions** or from the [Marketplace](#).

- [Analysis Services](#)
- [Integration Services](#)
- [Reporting Services](#)

SSDT for Visual Studio 2017

Changes in SSDT for Visual Studio 2017

Starting with Visual Studio 2017, the functionality of creating Database Projects has been integrated into the Visual Studio installation. There's no need to install the SSDT standalone installer for the core SSDT experience.

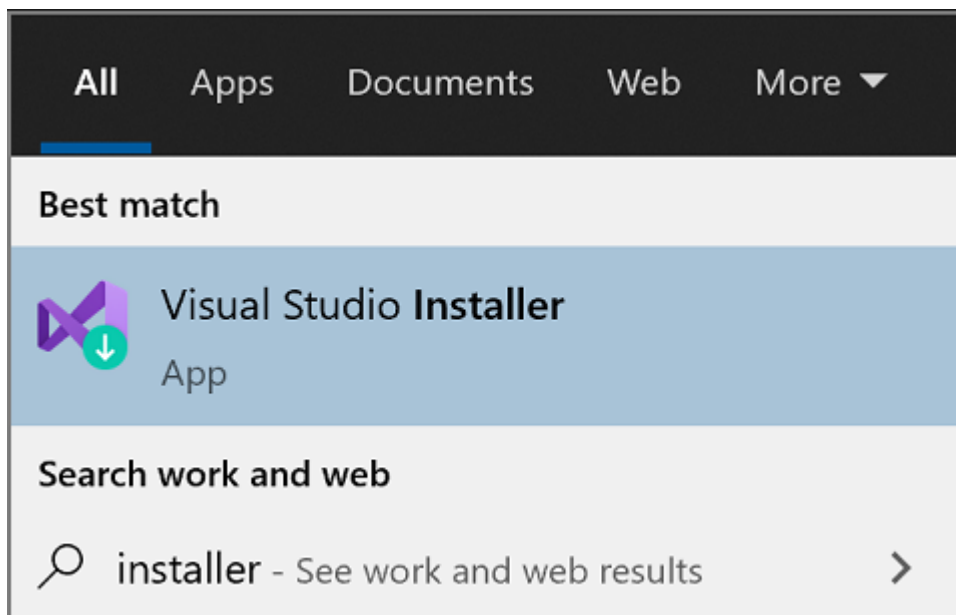
Now to create Analysis Services, Integration Services, or Reporting Services projects, you still need the SSDT standalone installer.

Install SSDT with Visual Studio 2017

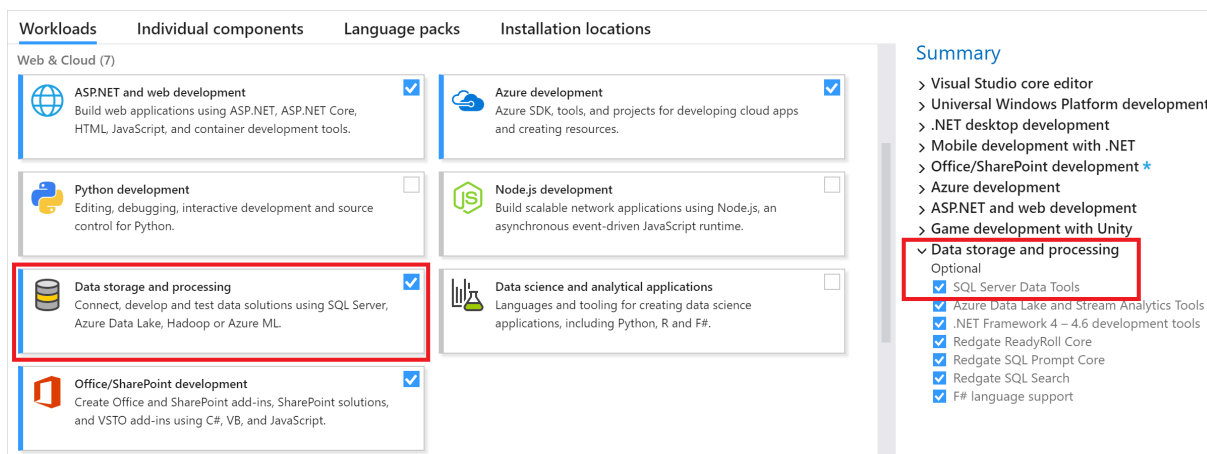
To install SSDT during [Visual Studio installation](#), select the **Data storage and processing** workload, and then select **SQL Server Data Tools**.

If Visual Studio is already installed, use the Visual Studio Installer to modify the installed workloads to include SSDT.

1. Launch the Visual Studio Installer. In the Windows Start menu, you can search for "installer".



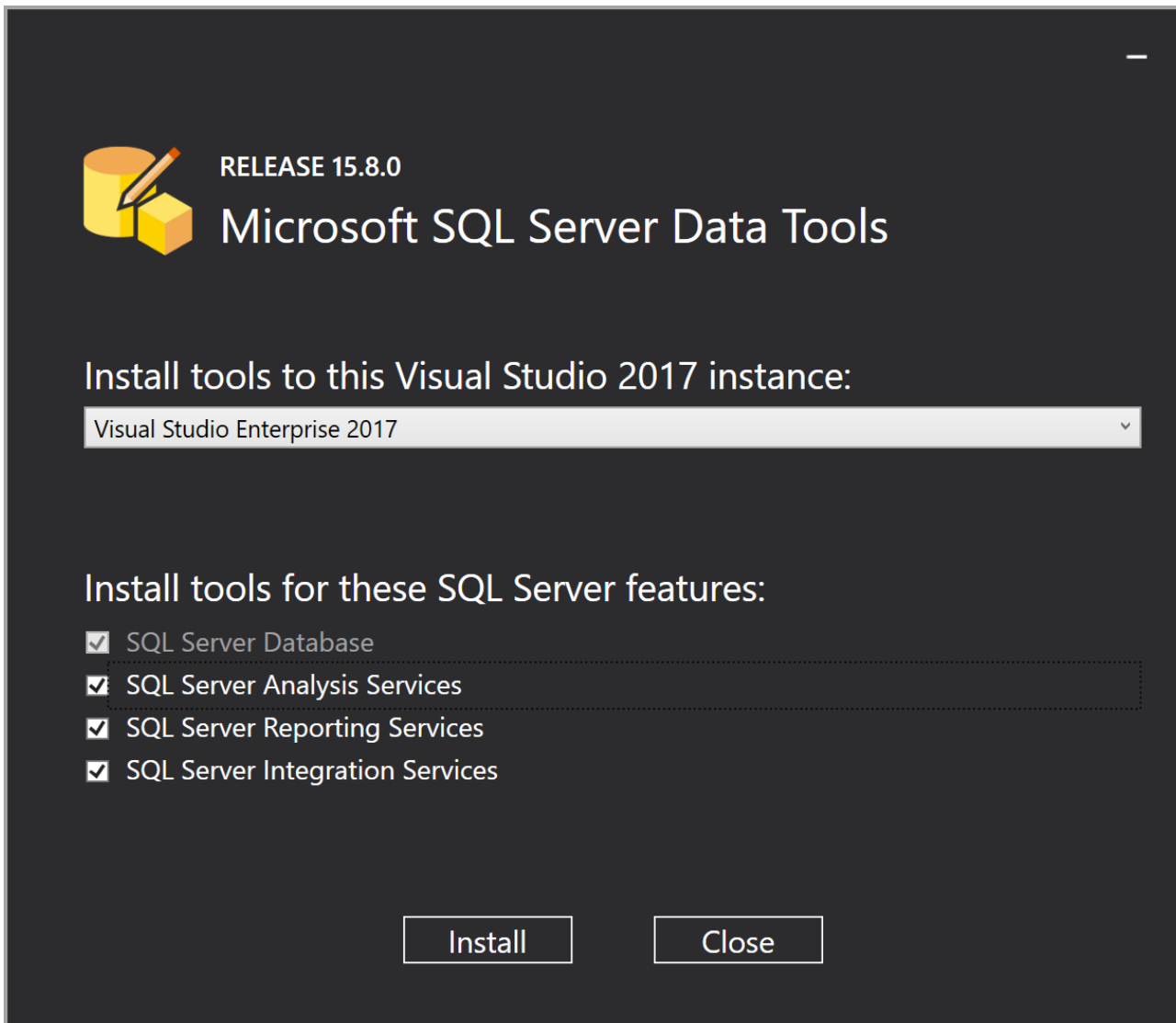
2. In the installer, select for the edition of Visual Studio that you want to add SSDT to, and then choose **Modify**.
3. Select **SQL Server Data Tools** under **Data storage and processing** in the list of workloads.



Install Analysis Services, Integration Services, and Reporting Services tools

To install Analysis Services, Integration Services, and Reporting Services project support, run the [SSDT standalone installer](#).

The installer lists available Visual Studio instances to add SSDT tools. If Visual Studio isn't already installed, selecting **Install a new SQL Server Data Tools instance** installs SSDT with a minimal version of Visual Studio, but for the best experience, we recommend using SSDT with [the latest version of Visual Studio](#).



SSDT for VS 2017 (standalone installer)



[Download SSDT for Visual Studio 2017 \(15.9.6\)](#)

i Important

- Before installing SSDT for Visual Studio 2017 (15.9.6), uninstall *Analysis Services Projects* and *Reporting Services Projects* extensions if they are already installed, and close all VS instances.

- Removed the inbox component Power Query Source for SQL Server 2017. Now we have announced Power Query Source for SQL Server 2017 & 2019 as out-of-box component, which can be downloaded [here](#).
- To design packages using Oracle and Teradata connectors and targeting an earlier version of SQL Server prior to SQL 2019, in addition to the **Microsoft Oracle Connector for SQL 2019** and **Microsoft Teradata Connector for SQL 2019**, you need to also install the corresponding version of Microsoft Connector for Oracle and Teradata by Attunity.
 - **Microsoft Connector Version 5.0 for Oracle and Teradata by Attunity targeting SQL Server 2017**
 - **Microsoft Connector Version 4.0 for Oracle and Teradata by Attunity targeting SQL Server 2016**
 - **Microsoft Connector Version 3.0 for Oracle and Teradata by Attunity targeting SQL Server 2014**
 - **Microsoft Connector Version 2.0 for Oracle and Teradata by Attunity targeting SQL Server 2012**

Release Notes

For a complete list of changes, see [Release notes for SQL Server Data Tools \(SSDT\)](#).

System requirements

SSDT for Visual Studio 2017 has the same [system requirements](#) as Visual Studio.

Available Languages - SSDT for VS 2017

This release of **SSDT for VS 2017** can be installed in the following languages:

- [Chinese \(Simplified\)](#)
- [Chinese \(Traditional\)](#)
- [English \(United States\)](#)
- [French](#)
- [German](#)
- [Italian](#)
- [Japanese](#)

- [Korean](#)
- [Portuguese \(Brazil\)](#)
- [Russian](#)
- [Spanish](#)

Considerations and limitations

- You can't install the community version offline
- To upgrade SSDT, you need to follow the same path used to install SSDT. For example, if you added SSDT using the VSIX extensions, then you must upgrade via the VSIX extensions. If you installed SSDT via a separate install, then you need to upgrade using that method.

Offline install

To install SSDT when you're not connected to the internet, follow the steps in this section. For more information, see [Create a network installation of Visual Studio 2017](#).

First, complete the following steps while **online**:

1. [Download the SSDT standalone installer](#).
2. [Download vs_sql.exe](#).
3. While still online, execute one of the following commands to download all the files required for installing offline. Using the `--layout` option is the key, it downloads the actual files for the offline installation. Replace `<filepath>` with the actual layouts path to save the files.
 - a. For a specific language, pass the locale: `vs_sql.exe --layout c:\<filepath> --lang en-us` (a single language is ~1 GB).
 - b. For all languages, omit the `--lang` argument: `vs_sql.exe --layout c:\<filepath>` (all languages are ~3.9 GB).

After completing the previous steps, the following steps below can be done **offline**:

1. Run `vs_setup.exe --NoWeb` to install the VS2017 Shell and SQL Server Data Project.
2. From the layouts folder, run `SSDT-Setup-ENU.exe /install` and select SSIS/SSRS/SSAS.
 - a. For an unattended installation, run `SSDT-Setup-ENU.exe /INSTALLALL[:vsinstances]`

/passive.

For available options, run `SSDT-Setup-ENU.exe /help`

 Note

If using a full version of Visual Studio 2017, create an offline folder for SSDT only, and run `SSDT-Setup-ENU.exe` from this newly created folder (don't add SSDT to another Visual Studio 2017 offline layout). If you add the SSDT layout to an existing Visual Studio offline layout, the necessary runtime (.exe) components are not created there.

Supported SQL versions

Project Templates	SQL Platforms Supported
Relational databases	SQL Server 2005* - SQL Server 2017 (use SSDT 17.x or SSDT for Visual Studio 2017 to connect to SQL Server on Linux) Azure SQL Database Azure Synapse Analytics (supports queries only; database projects aren't yet supported) * SQL Server 2005 support is deprecated, move to an officially supported SQL version
Analysis Services models	SQL Server 2008 - SQL Server 2017
Reporting Services reports	
Integration Services packages	SQL Server 2012 - SQL Server 2019

DacFx

SSDT for Visual Studio 2015 and 2017 both use DacFx 17.4.1: [Download Data-Tier Application Framework \(DacFx\) 17.4.1](#).

Previous versions

To download and install SSDT for Visual Studio 2015, or an older version of SSDT, see [Previous releases of SQL Server Data Tools \(SSDT and SSDT-BI\)](#).

See Also

- [SSDT MSDN Forum](#)
- [SSDT Team Blog](#)
- [DACFx API Reference](#)
- [Download SQL Server Management Studio \(SSMS\)](#)

Next steps

After installing SSDT, work through these tutorials to learn how to create databases, packages, data models, and reports using SSDT.

- [Project-Oriented Offline Database Development](#)
- [SSIS Tutorial: Create a Simple ETL Package](#)
- [Analysis Services tutorials](#)
- [Create a Basic Table Report \(SSRS Tutorial\)](#)

Get help

- [UserVoice: Have suggestions for improving SQL Server?](#)
- [Microsoft Q & A \(SQL Server\)](#)
- [DBA Stack Exchange \(tag sql-server\): Ask SQL Server questions](#)
- [Stack Overflow \(tag sql-server\): Answers to SQL development questions](#)
- [Reddit: General discussion about SQL Server](#)
- [Microsoft SQL Server License Terms and Information](#)
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- [Contact Microsoft](#)
- [Additional SQL Server help and feedback](#)

Is this page helpful?

 Yes  No
